

Nassau Candy Distributor

Shipping Route Efficiency Analysis

Executive Summary — Prepared for Unified Mentor

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Purpose of This Report

This executive summary presents the key findings and strategic recommendations from the Factory-to-Customer Shipping Route Efficiency Analysis conducted for Nassau Candy Distributor. The analysis covers 10,194 shipment records across 196 unique routes, four US regions, 59 states, and five manufacturing facilities. The objective is to provide leadership with clear, data-driven visibility into logistics performance — identifying bottlenecks, quantifying delay exposure, and recommending targeted interventions.

Business Performance Snapshot

10,194	1,321d	33.1%	66.9%	\$141.8K	\$93.4K
Total Orders	Avg Lead Time	Delay Rate	On-Time Rate	Total Revenue	Gross Profit

Situation Assessment

Nassau Candy Distributor ships from five factories — **Lot's O' Nuts** (AZ), **Wicked Choccy's** (GA), **Sugar Shack** (MN), **Secret Factory** (IL), and **The Other Factory** (TN) — to customers across the continental United States. Lead times range from **904 to 1,642 days**, a difference of 738 days between the best and worst performing routes. This gap represents a significant operational inefficiency that directly affects customer satisfaction and competitive positioning.

Critical Findings

■ Factory Performance Gap

The Other Factory (TN) leads with avg 1,280 days vs Sugar Shack at 1,340 days — a 60-day gap on the same logistics network. Sugar Shack's 48.5% delay rate is nearly 1.5× the company average of 33.1%.

■ Ship Mode Paradox

Standard Class delivers 24 days faster than First Class (1,314 vs 1,338 avg days). Premium shipping modes are not delivering faster — indicating operational mis-prioritization and potential overspend on expedited services.

■ ■ Pacific Region Bottleneck

The Pacific region has the highest delay rate (34.2%) and is the only region where delay rates exceed 34%. Interior region has the highest order volume (3,583 orders) with a 33.6% delay rate — the highest absolute delay count.

■ ■ Route-Level Risk Concentration

Ten routes have 100% delay rates, all hitting the maximum observed lead time ceiling of 1,638–1,642 days. These routes from Sugar Shack, Secret Factory, and Lot's O' Nuts to states including NJ, NH, CT, and WV represent the highest customer-satisfaction risk.

■ Bimodal Lead Time Distribution

Lead times cluster around two distinct bands: 904–950 days (fast pipeline) and 1,600–1,642 days (slow pipeline). The absence of a middle band suggests structural differences in carrier contracts or routing logic rather than random variation.

Strategic Recommendations

Priority	Action	Expected Impact
■ High	Audit & escalate 10 routes with 100% delay rates	Immediate customer satisfaction recovery
■ High	Investigate Sugar Shack supply chain vulnerabilities	Reduce 48.5% delay rate toward < 25%
■ Medium	Replicate The Other Factory logistics model company-wide	Potential 40-day reduction in avg lead time
■ Medium	Renegotiate Pacific region carrier contracts	Target < 30% delay rate in Pacific region
■ Low	Reassign First Class & Same Day to prioritized carriers	Justify premium shipping cost premium
■ Low	Establish regional hubs in Interior & Pacific	Long-term structural lead time reduction
■ Ongoing	Deploy Streamlit dashboard for continuous monitoring	Data-driven, proactive logistics decisions

Conclusion

This analysis establishes clear, data-driven visibility into Nassau Candy Distributor's shipping operations. The 738-day gap between best and worst performing routes, the 33.1% overall delay rate, and the underperformance of premium shipping modes represent significant opportunities for operational improvement. Implementing the high-priority recommendations above is projected to reduce the overall delay rate from 33.1% toward the 20% industry benchmark — improving customer experience, reducing logistics costs, and building a more scalable national distribution network.

The accompanying interactive Streamlit dashboard provides ongoing monitoring capability, enabling the logistics team to track route performance in real time and respond proactively to emerging bottlenecks.